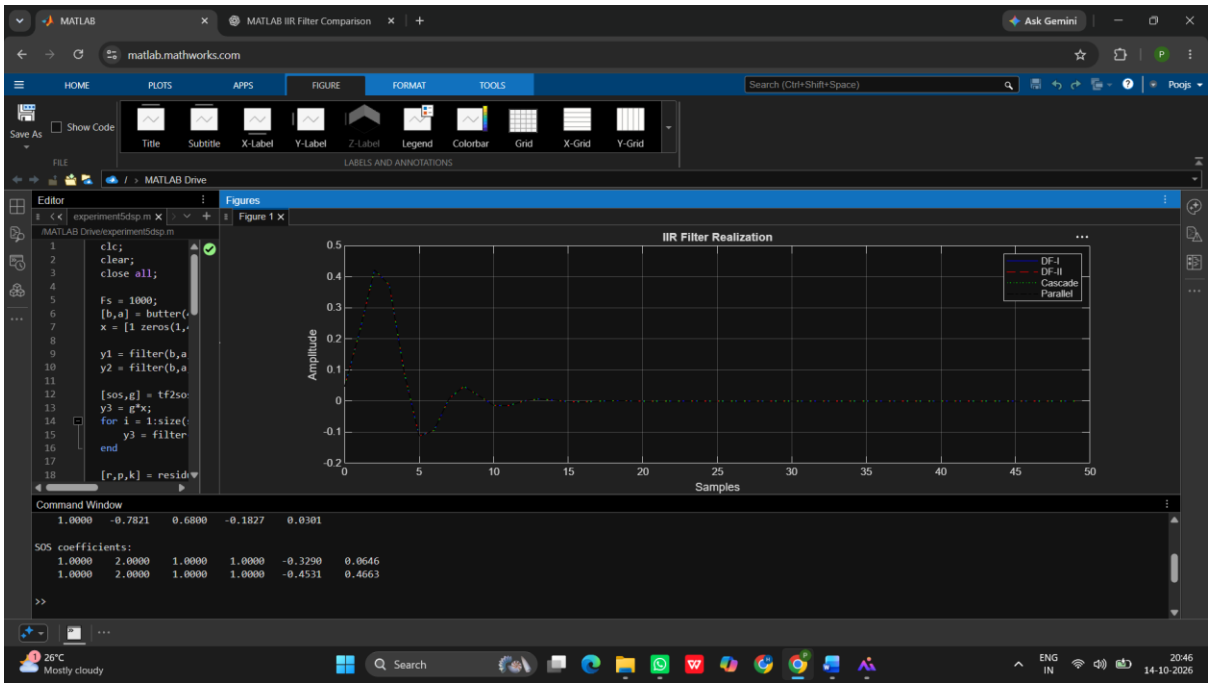
EXPERIMENT 5

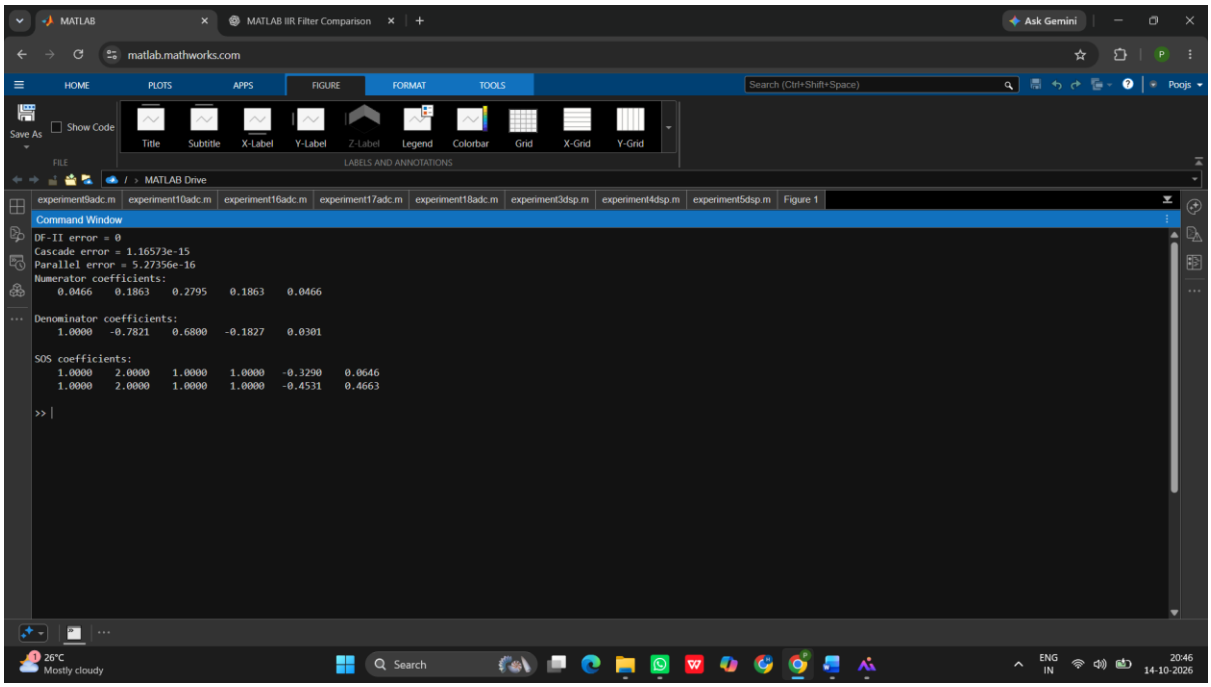
NAME: Pooja V

REG NO: 192512463

PROGRAM:

```
clc;
clear;
close all;
Fs = 1000;
[b,a] = butter(4,200/(Fs/2));
x = [1 zeros(1,49)];
y1 = filter(b,a,x);
y2 = filter(b,a,x);
...fprintf('DF-II error = %g\n',max(abs(y1-y2)));
fprintf('Cascade error = %g\n',max(abs(y1-y3)));
fprintf('Parallel error = %g\n',max(abs(y1-y4)));
disp('Numerator coefficients:');
disp(b);
disp('Denominator coefficients:');
disp(a);
disp('SOS coefficients:');
disp(sos);
```


MATLAB
MATLAB IIR Filter Comparison
Ask Gemini
matlab.mathworks.com
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Pooja
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Editor
Figures
Figure 1
IIR Filter Realization
DF-I
DF-II
Cascade
Parallel
Amplitude
Samples
Command Window
1.0000 -0.7821 0.6800 -0.1827 0.0301
SOS coefficients:
1.0000 2.0000 1.0000 1.0000 -0.3290 0.0646
1.0000 2.0000 1.0000 1.0000 -0.4531 0.4663
26°C
Mostly cloudy
Search
ENG
IN
20:46
14-10-2026


MATLAB
MATLAB IIR Filter Comparison
Ask Gemini
matlab.mathworks.com
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Pooja
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
experiment9adc.m
experiment10adc.m
experiment16adc.m
experiment17adc.m
experiment18adc.m
experiment3dsp.m
experiment4dsp.m
experiment5dsp.m
Figure 1
Command Window
DF-II error = 0
Cascade error = 1.16573e-15
Parallel error = 5.27356e-16
Numerator coefficients:
0.0466 0.1863 0.2795 0.1863 0.0466
Denominator coefficients:
1.0000 -0.7821 0.6800 -0.1827 0.0301
SOS coefficients:
1.0000 2.0000 1.0000 1.0000 -0.3290 0.0646
1.0000 2.0000 1.0000 1.0000 -0.4531 0.4663
26°C
Mostly cloudy
Search
ENG
IN
20:46
14-10-2026